035 Inventory - App Summary

What It Is

Operator-grade, desktop-first inventory, warehouse, and compliance system for industrial drones and supply chain assets, with English and Chinese support.

Who It Is For (Primary Persona)

Warehouse operator - scan-first warehouse workflows, exception handling, and cycle counts.

What It Does (Key Features)

- Scan-driven warehouse operations: receive, quarantine or release, put away, pick, pack, and ship with validation.
- Inventory traceability via append-only movement ledger and explainable changes.
- Compliance gating for operations with explicit missing or expired requirements.
- Lifecycle tracking for drones and batteries, including readiness and grounding states.
- Optional Kingdee Cloud finance sync with mappings and reconciliation.

How It Works (Architecture Overview)

- Frontend: Dioxus 0.7.x desktop-first UI with Tailwind; selective mobile subset and a web build for testing.
- Backend: Rust Axum server exposes GraphQL; runs migrations on startup and enables CORS for web UI access.
- Persistence: SQLx with SQLite database pool; default database URL is sqlite:data/drone-inventory.db.
- CLI: Rust clap tooling for database migrations, backup and restore, seed import, compliance utilities, and integrations.

Data Flow (Observed)

- Dioxus UI (desktop or web) sends requests to the GraphQL API.
- GraphQL handlers use backend services and repositories over SQLx.
- SQLite stores operational data; migrations run at server start.
- Configuration loads from optional config file plus APP__* environment variables.

How To Run (Minimal Getting Started)

- Start the backend GraphQL server: ./scripts/run-server.sh (defaults to 127.0.0.1:3000, runs migrations).
- Install UI build deps for Tailwind (Node 18+): cd assets/ui && npm install.
- Run the web UI: ./scripts/run-web.sh (builds Tailwind CSS, then runs dx serve --platform web --port 8080).

Not Found In Repo

- Explicit installation steps for Rust toolchain and the Dioxus CLI (dx).
- Production deployment or hosting instructions.